

CSE 8803: Deep Learning for Text Data

Fall 2024

Homework 2

Due: Oct.10, 2024

1. **(30 points) Scalable Inference**

In the context of large language models, decoding refers to the process of generating sequences of words or tokens based on the input provided to the model. At each iteration or time step, the model uses the previously generated token or word to predict the next most probable token from the vocabulary. Speculative Decoding [4] is a method to improve LLM inference speed by around $2-3\times$ without degrading any accuracy. Please answer the following questions.

- (a) (5 points) (True/False) In speculative decoding, the target model and the draft model must both use the same tokenizer. Please also explain your answer.
- (b) (10 points) Please summarize what speculative decoding is and how it differs from other decoding methods like greedy decoding or beam search.
- (c) (15 points) A common technique for enhancing the performance of large model inferences is leveraging the Key-Value (KV) cache from previous inferences. Utilizing this cache boosts inference speed and reduces end-to-end latency, all without compromising accuracy. Please read the paper [8] and explain why we only use KV cache but do not cache the query (Q) vectors.

Answer:

(a)

True. Because we need to use target model M_p output and probability distribution to evaluate draft model M_q parallel. If tokenizers are different, the comparison and draft would make no sense.

(b)

- **What speculative decoding is:** Speculative decoding is a method and algorithm to accelerate language model inference by parallelly computing tokens. It uses a draft model to generate rough token candidate, and target model verify or update the token set.
- **How it differs from other decoding method:**
 - It may be faster and more accurate than other decoding method.
 - Greedy search decoding could be less accurate than speculative decoding. Greedy Search Encoding selects the local optimal word in each time step or iteration, but it could not get the words later predicated, leading to worse prediction accuracy for the full sentence(global solution).
 - Beam search adopted breath-first search tree. Therefore, being faster means to reduce beam widths and sacrifice accuracy. If beam width is one, it becomes greedy search[7].

(c)

Because KV cache too many tokens, using more memory than one token per example of Q, K, V in orders of magnitude.

- Q is generated one per token and would not influence computation bottleneck drastically.
- Q tensors are very small compared to KV , so it's profitable to allocate all-to-all communication time to Q .

2. (40 Points) Pre-Trained Language Models

In the lecture, we have learned that Transformer-based pre-trained language models (PLMs) can be categorized into three types: 1) Encoder-based PLMs such as BERT [2] and RoBERTa [6]; 2) Decoder-based PLMs such as GPT [1] and LLaMA [11]; 3) Encoder-Decoder PLMs such as BART [5] and T5 [9].

- (5 points) Please explain how each architecture processes input sequences.
- (5 points) For all three architectures (Encoder-only, Encoder-Decoder, Decoder-only), please propose an NLP task that each is best suited for and justify your reasoning (e.g., translation, summarization, question-answering, etc.)

In this question, you are asked to draw three figures that illustrate how these three categories of PLMs are applied to different tasks.

- Using BERT for binary sentiment analysis:** Classify whether the sentiment is positive or negative for the sentence “I’ve never been so utterly blown away, yet simultaneously so unbelievably frustrated by a game as I have been with Black Myth: Wukong” [10].
- Using T5 for machine translation:** Translate the following sentence to German: “The house is wonderful.” [3].
- Using Llama-3-Instruct for machine translation:** Translate the following sentence to English: “Je suis étudiant.” or “Je suis étudiante.” (select either one).

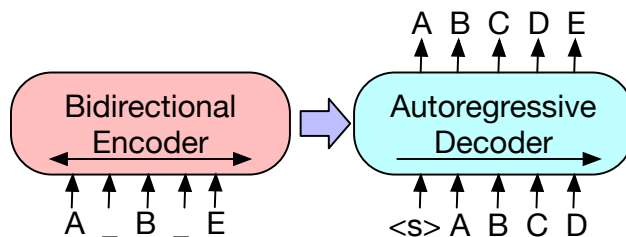


Figure 1. Example figure, from [5]

Requirements

- The figures must be in PDF or another vector graphics format. Bitmap or hand-drawn figures will receive 0 points.
- You do not need to draw the architectural details within the Transformer blocks.
- You can treat each word as a token. Punctuation marks should be treated as individual tokens.
- For the sentiment analysis task, you may omit some words within the sentence and represent the sentence as “I’ve never been ... Black Myth: Wukong”. The output should be post-activation but pre-ArgMax.
- The input tokens should be in black, and all model-generated tokens or labels should be in blue.

- You should show clear correspondence between input tokens and model-generated tokens or labels.
- For the machine translation tasks, you need to draw the entire input and output sequences.

Figure 1 shows an example. Note that it does not meet all the requirements.

(a)

- **Encoder-based PLMs(such as BERT):** Only use bidirectional encoder, considering the context is both left to right and left to right directions.
- **Decoder-based PLMs (such as GPT and LLaMA):** Only use autoregressive decoder.
- **Encoder-Decoder PLMs(such as T5):** Use bidirectional encoder to process input sequence, and output to autoregressive decoder. Autoregressive generate tokens one by one.

(b)

- **Encoder-only PLMs - Sentiment Analysis** Because these models can capture full semantics in both text direction, analysis result will be based on more comprehensive and nuanced understanding.
- **Decoder-only PLMs - Text Generation/ Question-Answering** Because these models generate text in sequence based on previous token, they have better ability to output proper result based on prior text.
- **Encoder-Decoder PLMs - Machine Translation** Because machine translation is a sequence to sequence task, which is suitable for encoder-decoder method.

(c) Please find in the next page.

3. (30 points) BERT for Named Entity Recognition

Please check the GitHub repository for this coding assignment: <https://github.com/Yinghao-Li/CSE8803DLT-24Fall-Assignment2-Public>.

This assignment contains 5 code blocks to complete, each with a 6-point value.

References

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- [3] huggingface. T5 — huggingface.co. https://huggingface.co/docs/transformers/en/model_doc/t5.
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- [5] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. BART: denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel R. Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, ACL 2020, Online, July 5-10, 2020*, pages 7871–7880. Association for Computational Linguistics, 2020.
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- [7] Matt Payne. What is beam search? explaining the beam search algorithm, September 2021.
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- [10] Mitchell Saltzman. Black Myth: Wukong Review - IGN — ign.com. <https://www.ign.com/articles/black-myth-wukong-review-pc>.

- [11] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurélien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models. *CoRR*, abs/2302.13971, 2023.